

V. CONCLUSION

We have derived a series representation for the Green's dyadics in a bianisotropic medium with material dyadics of the form $\bar{\bar{\mu}} = \alpha \bar{\bar{\epsilon}}^T$, $\bar{\bar{\xi}} = \xi \mathbf{u}_z \mathbf{u}_z$ and $\bar{\bar{\zeta}} = \zeta \mathbf{u}_z \mathbf{u}_z$. When the condition $2(\bar{\bar{\mu}} + \bar{\bar{\mu}}^T)^{-1} : \mathbf{u}_z \mathbf{u}_z = \bar{\bar{\mu}}^{-1} : \mathbf{u}_z \mathbf{u}_z$ is satisfied it was possible to derive a closed form representation.

At present the factorization of the Helmholtz operator and sometimes the closed-form Green's dyadics are known for different classes of media. These classes are not necessarily generalizations of each other such as for example the medium of this paper and the one studied in [4]. A challenge for future research is finding a medium that allows factorization and that contains all the presently studied media as special cases.

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Neural Network-Based CAD Model for the Design of Square-Patch Antennas

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Abstract—An artificial neural network (ANN) has been developed and tested for square-patch antenna design. It transforms the data containing the dielectric constant (ϵ_r), thickness of the substrate (h), and antenna's dominant-mode resonant frequency (f_r) to the patch length (l).

Index Terms—CAD, neural network, patch antenna.

I. INTRODUCTION

Recently, nonlinear neural optimization networks have been applied [1] for analysis of arbitrarily shaped microstrip patch antennas with a very general bianisotropic grounded slab. On design side, reported works [2] are limited. In the design of square-patch antennas, the resonant frequency in the dominant mode is given, ϵ_r and h are supplied, and l is calculated. In this letter, the authors extend the work [3] on the use of the artificial neural network (ANN) technique in place of conventional numerical techniques, for the design.

II. DATA GENERATION AND PREPROCESSING

The resonant frequency of a square-patch antenna in its dominant mode is given by Wolf and Knoppik [4] as

$$f_r = \frac{c}{2l_{\text{eff}} \sqrt{\epsilon_{\text{dyn}}}} \quad (1)$$

where c is the velocity of light in free-space, l_{eff} is the effective patch length (which is a function of l and h), ϵ_{dyn} is the dynamic dielectric constant considering the fringing field effects (which is also a function of l , h , and ϵ_r).

Determination of l from (1) is not direct. Mathematical manipulations express it as an iterative equation, which is to be solved by standard numerical techniques. ANN technique reduces the numerical and computational complexities.

Equation (1) is used to generate data for $1.17 \leq \epsilon_r \leq 15.0$, $0.0195 \text{ cm} \leq h \leq 0.3125 \text{ cm}$, and $0.2 \text{ cm} \leq l \leq 10.0 \text{ cm}$. The generated data were then arranged in two matrices. A three-column matrix containing the values of ϵ_r , h , and f_r and is used as the input matrix to the network. The other one is a column matrix containing the corresponding values of l . The elements of the second matrix are then normalized with respect to highest value of l (i.e., 10) for transforming their range to $[0, 1]$ during training [5]. This is used as the output matrix.

III. APPLYING NEUROCOMPUTING TECHNIQUE

In the present work, we have adopted the same optimization strategy that is used for error back propagation algorithm [6] in conjunction with the generalized criterion proposed by Karayiannis *et al.* [5] for training of a three-layer neural network (i.e., single hidden layer). The algorithms are based on the minimization of generalized

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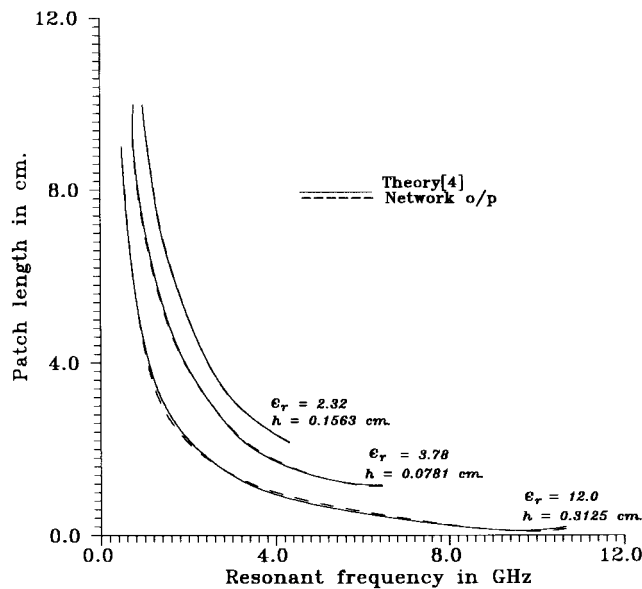


Fig. 1. Comparison of network output with theory [4].

objective function $G(\lambda) = \lambda E + (1 - \lambda)E'$, where E is the mean-square error function. λ changes during the training according to the rule $\lambda = \lambda(E) = \exp(-\mu/E^2)$.

Out of the 1290 data generated, 645 were used for training and the rest were used for testing of the trained neural network. The optimized values of different parameters, which are obtained by trial and error for the training of the network, are: 1) the number of input nodes = 3; 2) the number of neurons in the hidden layer = 40; 3) the number of output node = 1; 4) the learning rate parameter (α) = 0.002; 5) the learning rate adaption (μ) = 1.5; and 6) momentum (β) = 5.0.

IV. RESULTS

Network testing was performed both for theoretical and experimental data. Fig. 1 demonstrates the comparison between theoretical patch length and that obtained using the developed ANN model. A square-patch antenna with $l = 4.14$ cm, $\epsilon_r = 2.5$, $h = 0.1524$ cm is experimentally found to resonate at 2.228 GHz. When the developed network model is used for the same antenna, the patch length is found to be 4.151 cm.

V. DISCUSSION

A distinct advantage of neurocomputing is that, after proper training, a neural network completely bypasses the repeated use of complex iterative processes for new cases presented to it. Our model produces the length of the side of a square patch almost instantaneously for the other three specified parameters of the square-patch antenna. The training time on an HP 9000/712 workstation platform is 13 min. Besides that, this single network structure can predict the results for frequency range up to 15 GHz provided the values of ϵ_r and h are in the domain of training values.

VI. CONCLUSION

A neural network-based CAD model is developed for the design of a square-patch antenna, which is robust both from the angle of time of computation and accuracy.

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Neural Network-Based Adaptive Beamforming for One- and Two-Dimensional Antenna Arrays

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Abstract—In this letter, we present a neural network approach to the problem of finding the weights of one- (1-D) and two-dimensional (2-D) adaptive arrays. In modern cellular satellite mobile communications systems and in global positioning systems (GPS's), both desired and interfering signals change their directions continuously. Therefore, a fast tracking system is needed to constantly track the users and then adapt the radiation pattern of the antenna to direct multiple narrow beams to desired users and nulls interfering sources. In the approach suggested in this paper, the computation of the optimum weights is accomplished using three-layer radial basis function neural networks (RBFNN). The results obtained from this network are in excellent agreement with the Wiener solution.

Index Terms—Adaptive arrays, antenna arrays, beamforming, neural network applications, tracking.

I. INTRODUCTION

Neural networks are gaining momentum in the field of signal processing [1], [2] mainly because of their general-purpose nature, fast convergence rates, and new very large scale integration (VLSI) implementations. Motivated by these inherent advantages, this paper presents the development of a neural network-based algorithm to compute the weights of an adaptive array antenna [3]. In this new

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